| Date | 8 July 2026 | Project Name | Heart Disease Analysis |
|---|---|---|---|
| Team ID | — | Maximum Marks | — |

# PROJECT PERFORMANCE TESTING

## Model Performance Testing & Dashboard Implementation

**Team Leader:** Mounika Sanapathi
**Team Members:** Thakasi Balaji, Kale Sai Praneeth, Navyasri Gonnuri, Shanthi Kumari Challa

| S.No. | Parameter | Values / Implementation Details |
|---|---|---|
| 1. | **Data Rendered** | The clinical heart disease dataset was successfully uploaded and rendered in Tableau Public. All key dataset fields—including *Age, Sex, Chest Pain Type (cp), Resting Blood Pressure (trestbps), Serum Cholesterol (chol), Fasting Blood Sugar (fbs), Resting Electrocardiographic Results (restecg), Maximum Heart Rate Achieved (thalach), Exercise Induced Angina (exang), ST Depression (oldpeak)*, and the target diagnostic field—were loaded correctly and made available for analytical mapping. |
| 2. | **Data Preprocessing** | Removed missing or null physiological metrics, verified medical data schema and variable formats, initialized custom calculated diagnosis fields, grouped patient cholesterol profiles, and validated continuous metric bounds before rendering primary visual worksheets. |
| 3. | **Utilization of Filters** | **Aggregation Functions:** `COUNT(Patient_ID)` was primarily implemented for baseline cohort counts, and `AVG(thalach)` or `AVG(chol)` parameters were applied to track metric variations across condition categories.<br><br>**Action Filters:** Worksheets utilize coordinated, dynamic target action filters to achieve synchronized dashboard-wide patient profiles cross-filtering upon selecting specific age cohorts or risk categories. |
| 4. | **Calculation Fields Used** | Created interactive calculated fields to isolate total metrics (e.g., Target Patients Segment Risk Profile).<br><br>**Logic / Syntax:** Mapped via custom physiological risk integer boundaries linked with index functions:<br>`INDEX()`<br>This approach filters records based on selection rankings.<br><br>**Purpose:** Solves Tableau's order of operations sequence constraint, |

| S.No. | Parameter | Values / Implementation Details |
|---|---|---|
| | | enabling the clinical dashboard to isolate and rank high-risk patient subgroups dynamically within specified age-bracket regions without breaking structural constraints. |
| **5.** | **Dashboard Design** | **All Visualizations Unified into Dashboard Design (Heart Disease Analysis Dashboard):**<br>• **Total Patient Records:** 50,000 profiles evaluated.<br>• **Primary Metric Distributions:** Maximum heart rate distributions across health thresholds.<br>• **Cuisine**/**Demographic Slices:** Patient segmentation charts by gender brackets.<br>• **Heatmaps** / **Correlations:** Risk matrix correlating ST depression markers and cholesterol thresholds. |
| **6.** | **Story Design** | **Heart Disease Analysis Tableau Story:** Multiple interactive worksheets and dashboards are layered into the unified Tableau story sequence (e.g., tracking clinical metrics from patient entry profiles to final cardiovascular risk analysis). |